

CHAPTER 7

INTERNET OF THINGS (IOT) APPLICATIONS IN LOGISTICS AND SUPPLY CHAIN MANAGEMENT

Mehmet Ali ERTÜRK*

*İstanbul University, School of Transportation and Logistics, İstanbul, Türkiye
E-mail: mehmetali.erturk@istanbul.edu.tr

DOI: 10.26650/B/SS05ET50.2024.016.07

ABSTRACT

Wireless connectivity has revolutionized all industries today by adopting the Internet of Things (IoT) technology stack. IoT enables any physical entity to be connected to the Internet and perform intelligent actions based on its environment. IoT-equipped sensors qualify logistics with real-time asset tracking, condition monitoring, data visualization, safety and security, fleet management, and other logistics and supply chain forecasting applications.

First, this study will provide general information about the Internet of Things architecture structure, data transmission infrastructure, and protocols. Then, the latest technological applications in logistics and supply chain management, such as artificial intelligence, blockchain, and big data digital twins, which are a part of the Internet of Things, will be discussed. The last section of this study will guide logistics and supply chain management with IoT by discussing the gaps in future projections and work areas for academia and industry.

Keywords: IoT, Supply Chain, Smart Logistic

1. Introduction

The widespread use of the Internet, the decrease in hardware costs, and the developments in the field of artificial intelligence have allowed the development of the Internet and the emergence of the concepts of the Internet of Things. In particular, the easier accessibility of wireless communication technologies and the decrease in unit costs have started a new digitalization process in many industries. This study will discuss general information about the essential components of the Internet of Things and its applications in logistics and supply chain management.

We can define the Internet of Things (IoT) as structures where objects are connected to the Internet with wireless communication technology and contain smart sensors. The approach in IoT allows sensors and smart devices to perform necessary tasks without any human interaction. Apart from connecting objects to the Internet and sending data, it is also aimed that these objects can be accessed and controlled over the Internet. Therefore, it can define the Internet of Things as an extensive global network that can perform specific tasks by communicating with each other and over the Internet of objects (1).

We can interact with the virtual world using physical objects thanks to the IoT. These devices can intelligently process the data they perceive or create, thanks to the sensors they contain or the customized functions. In this way, it enables us to exchange data over the Internet and develop smart applications. However, sensors that connect the physical world and the virtual environment have limited resources due to their nature. An IoT sensor has a limited processor, memory, hard disk, and battery, and it is impossible to perform all the work on the device. Instead, the collected data is transferred to the cloud environment, sometimes directly, sometimes with limited processing. In this way, tasks that bring more resources and computational capabilities (such as artificial intelligence applications) run on systems that can scale dynamically in cloud environments (2).

Thanks to the potential of the Internet of things, it is actively used in social, economic, health, education, military, logistics, engineering, intelligent power grids, home and entertainment systems, agriculture, etc., and is becoming widespread day by day (3). The widespread use of technology in various fields has led to digital transformation. Now, devices and sensors can generate data besides humans. The rapid growth of data has led to the transition of each industry to a data-driven business model, eventually leading to the emergence of application areas such as artificial intelligence, big data, etc. Sensors can record images, sound, or any signal, and these data can be analyzed with different methods to generate business value. The

inadequacy of classical techniques in data processing has led to the rapid development of artificial intelligence and data analytics. Industries that enable quick adaptation to artificial intelligence applications, such as customer segmentation, profitability planning, investment planning, and future forecasting, monetized the sector.

In the Internet of Things ecosystem, data is essential. There are different approaches to managing data, from centralized to decentralized management. Decentralized management involves multiple authorities reaching a consensus and is vital advancement in academia and industry. The blockchain is the most promising approach for decentralized approaches with the help of cryptographic tools to manage data securely and transparently. The technology has become popular with cryptocurrencies. However, its applications are not limited and can be used to track goods and services in logistics and supply chain use cases.

The rapid and real-time data transmission and developments in artificial intelligence have begun to change simulation approaches. These changes have started to be replaced by digital twins realized with real-time digital copy instead of simulations performed with abstraction methods. It is possible to perform real-time analysis and investigate what-if simulation scenarios through digital twins in logistics and supply chain applications.

The primary objective of this chapter is to introduce how next-generation information technologies contribute to logistics and supply chain management from a technology perspective. These technologies are the IoT and the relationships of the IoT with artificial intelligence, data analytics, blockchain, and digital twins from the logistics supply chain management standpoint.

In this chapter, the Second Section discusses the Internet of Things, data transmission, and application protocols. The Third Section demonstrates the IoT-based artificial intelligence, big data analytics, blockchain, and digital twin in logistics and supply chain management introduced. The final Section concludes the chapter.

2. Internet of Things (IoT)

Many building blocks comprise the Internet of Things, and it is impossible to discuss a single generally accepted architecture. Considering the applications, devices, and the Internet that make up the Internet of things, it is understood that it should have a multi-layered architecture. While three- and five-tier approaches stand out, architectures are not limited to this. The layers of the three and five approaches in Figure 1 are visible. If we inspect the three-layer structure:

- i. The perception layer: where data from different sensors are collected, and the physical environment is interacted with in this layer.
- ii. Network layer: the transmission of sensor data is also the data transmission layer, which includes one- or two-way designs.
- iii. The application layer is the product-oriented layer, and services are carried out through this layer.

The difference between the five-layer structure is that the transport layer refers to the layer that transmits the data, and the processing layer refers to the layer where the analysis is carried out, and the data is processed. The business layer is the layer of the business where the entire IoT ecosystem is run (2). We can discuss the building blocks of the IOT.

2.1. Data Communications in IoT

One of the Internet of Things’ essential components is physical sensors’ data communication. The highlight here is the mobility of devices thanks to wireless communication and the ability to communicate independently of the location. For this, first of all, the necessary infrastructure and the sensors must be capable of transmitting data from this layer. Many technologies serve this infrastructure today. At the beginning of these, LoRaWAN, LoRa, RFID, LTE, Z-Wave, NFC, UWB, M2M, 6LoWPAN, 5G, Nb-IoT, SigFox, IEEE 802.11, Bluetooth protocols are coming.

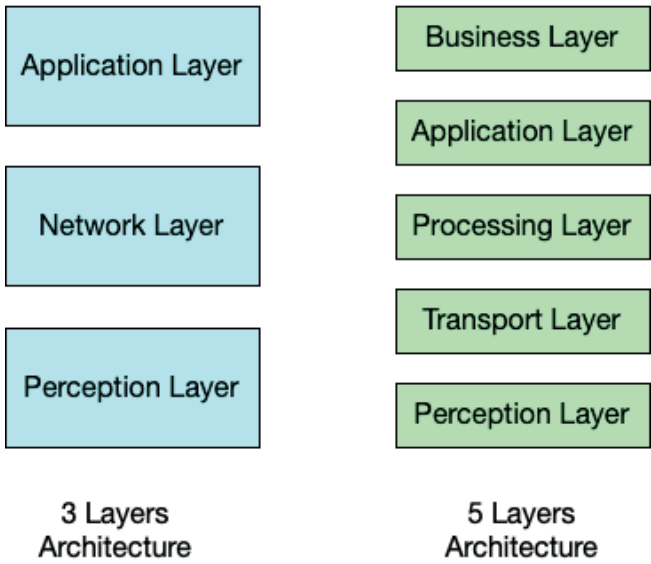


Figure 1. IoT Architecture Approaches (2)

Table 1: IoT Data Communications Technology Comparison						
	Sigfox	NB-IoT	LoRaWAN	WiFi	ZigBee	Bluetooth
Standards	Sigfox	3GPP	LoRa Alliance	IEEE 802.11	IEEE 802.15.4	Bluetooth SIG
Modulation	BPSK	QPSK	CSS	DSSS, OFDM	DSSS, QPSK	GFSK
Frequencies	ISM: 433 MHz, 868 MHz, 915 MHz	Licensed under LTE	ISM: 433 MHz, 868 MHz, 915 MHz	ISM: 2.4 GHz, 5 GHz	ISM: 868 MHz, 2.4 GHz	2.4 GHz
Coverage	10–40 km	2–20 km	1–10 km	10–100 m	10–100 m	10–100 m
Bandwidth	100 Hz	200 kHz	125 kHz, 250 kHz	20 MHz, 40 MHz, 80 MHz, 160 MHz	2 MHz	1 MHz
TX Limit	140 Packets per Day	Unlimited	Duty Cycle Lim.	Unlimited	Unlimited	Unlimited
Max Data Rate	100 bps	200 kbps	50 kbps	Gbps	250 kbs at 2.4 GHz	2 Mbps
Private Deployments	No	No	Yes	Yes	Yes	Yes
Energy Consumption	Low	Low	Low	High	Low	Low
Security	Low	High	High	Low-High	High	Low-High

Each technology has its communication structure, and there are differences according to the usage area. The prominent comparison of LoRaWAN, Sigfox, NB-IoT, WiFi, Zigbee, and Bluetooth is given in Table 1 (4). When compared, LoRaWAN works on an unlicensed spectrum; it can be deployed as public or private and enables long-distance data transmission with a builtin security model.

Although we do not have the opportunity to examine all communication technologies, we will evaluate the open protocol LoRaWAN, which is on its way to becoming a de facto standard today. In addition, we will mention that LoRaWAN comes to the fore in supply chain applications, among other communication networks.

LoRa stands for Long Range, a Radio Frequency (RF) modulation corresponding to the OSI reference model's physical layer. Whereas LoRaWAN is a MAC layer standard that coordinates the medium. In this section, we will discuss the technical aspects of both LoRa and LoRaWAN in detail. LoRaWAN is a MAC layer protocol that aims to solve medium and network congestion management issues. Any node using the LoRaWAN protocol can benefit from the following features provided by the standard; management of the medium, power consumption, adaptive data rates, and security(4).

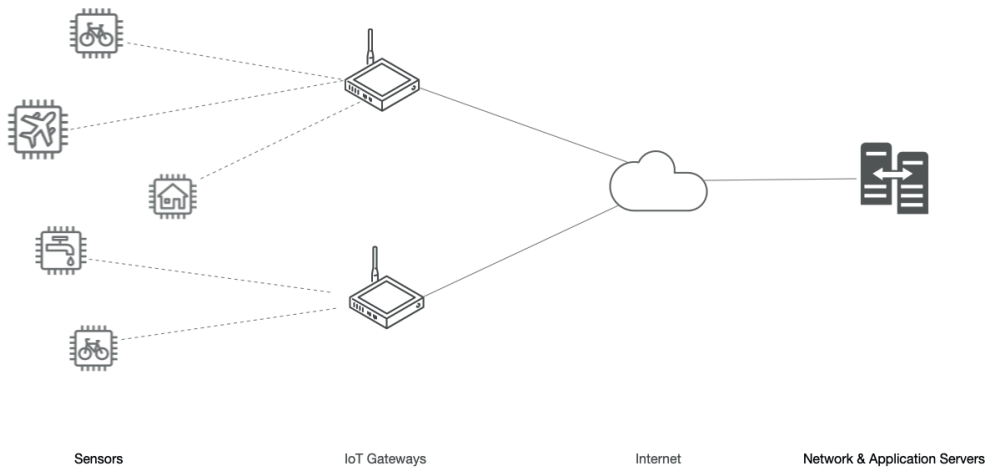


Figure 2. LoRaWAN Architecture (4)

A general architecture of LoRaWAN is given in Figure 2. The sensors can provide data communication up to 2 km in the dense urban and up to 15 km in rural areas with LoRa modulation. The data sent from the sensors is captured by the LoRaWAN base stations and routed to the network server. The network server manages the LoRaWAN network and forwards incoming messages to the corresponding application server. LoRaWAN is Low Power Wide Area Network (LPWAN) providing long-distance data transmission with low power consumption. This feature makes LoRaWAN an effective candidate for logistics and supply chain applications.

2.2. Application Protocols in IoT

Layer 7 of the OSI reference model is the application layer. For example, protocols that enable data exchange between computers on the Internet or in a local network are application layer protocols in the seventh layer.

Today, we use these applications, which have developed many application layers, daily. In the Internet of Things paradigm, the primary element is to provide an internet connection for devices with limited capacity and to provide applications and services with the data obtained from these devices. Therefore, many protocols have been developed in the application layer specialized for IoT.

One of these application protocols is the Constrained Application Protocol (CoAP). Designed with energy efficiency in mind, CoAP works over the UDP protocol. UDP offers a faster connection, so no session management is found in TCP. In addition, CoAP provides security using Datagram Transport Layer Security (DTLS) (5).

Another important IoT application protocol is Message Queue Telemetry Transporter (MQTT). The protocol works on the subscription principle. Parties subscribed to a particular topic are notified when a message is received on that topic. It is an effective method to transmit data, especially to devices with limited hardware resources. Since data transmission is in the form of broadcasts, bandwidth is used effectively (5).

An Extensible Markup Language (XML) complete protocol used in instant messaging applications is Extensible Messaging and Presence Protocol (XMPP). Advanced Message Queuing Protocol (AMQP) is a messaging protocol that works with subscription logic similar to XMPP. The working principles of AMQP are similar to MQTT (5).

3. Applications in Logistics and Supplychain

The age of smart production started after the fourth industrial revolution, and the essential tool of this age has been the Internet of things. It is the Industrial Internet of Things (IIoT), which is specialized for industry and on which Edustri 4.0 is emphasized. The fact that information technologies and production robots have become smart has become a part of the fact that the data obtained with IIoT, thanks to artificial intelligence, is carried out with lower production costs and personalized intelligent elements. With the expansion of IIoT, smart applications have started to take place in many sectors. Smart cities, smart transportation, smart health, smart living spaces, smart vehicles (Internet of Vehicles - IoV), autonomous vehicles, and big data applications are some of the prominent ones (6).

The Supply Chain encompasses all of the assets and processes that encompass orders, supplier to customer. SC process includes the right item to be delivered at the right price and the right volumes on time. This process can be very complex. However, we can overcome addressed issues using information technology and smart techniques. In this case, IIoT is the critical technology. Each product or service order can be tracked from supplier to customer with smart devices in real-time. In traditional supply chain processes, we need help delivering products on time or keeping excess stock. Using IIoT makes it possible to run supply chain processes more intelligent, connected, autonomously, integrated, and innovatively (7).

Making supply chain processes smart is possible with infrastructures to be designed in different areas. The first step is to make things smart. In this step, it will be ensured that all kinds of objects moving from one point to another in the procurement processes can be tracked. Different technologies can be used to make objects smarter. The first step is to make things smart to enable sensing, which corresponds to the perception layer. In this step, it will be ensured that all kinds of objects moving from one point to another in the procurement

processes can be tracked. Different technologies can be used to make objects smarter. The first of these is to associate the object with being tracked with sensors that provide long-range communication, such as RFID, Bluetooth, QR Codes, or LoRaWAN.

It will be shaped in itself in other processes according to the connection structure of the communication integrated attached to the objects. For example, in a warehouse where RFID is used, it is possible to read every object entering and leaving the warehouse with RFID readers and to follow the necessary movements. This approach will not only be the tracking of objects, but it will also be a part of the infrastructure that can trigger processes such as automatic stock counting and the ordering of new products under a certain stock or the supply of materials needed on the production line.

Equipping objects with smart sensors will improve tracking processes and learn and monitor the status of products with special needs, such as cold chains or healthcare applications. For example, these sensors can monitor the factors such as temperature, pressure, and leakage of the object they are in, generate a warning against possible danger, or inform the changes in the physical nature of the product. We can give a few examples of life-threatening gas leaks, monitoring the validity period of drugs, and physical changes such as cracks or fractures. This ability transforms physical entities into perceptible objects.

Intelligent sensors are generally limited in battery life, processor capacity, and memory resources. Therefore, the operations that can be performed on devices with limited resources are limited, and it is aimed to obtain maximum efficiency with the least possible operation. These sensors are not general purposes; they are just specialized tools for a specific target. So, algorithms running on the sensors are expected to be efficient regarding time and space complexity.

Today, energy-efficient operations can be performed on sensors with limited resources. Therefore, tools and methods that prioritize energy efficiency have been developed while designing algorithms and embedded software on sensors. The LoRaWAN protocol we mentioned in the previous section is one of them. It has methods and structures developed for wireless data transmission to provide energy efficiency in A, B, and C classes. The LoRaWAN protocol we mentioned in the previous section is one of them. It has methods and structures developed for wireless data transmission to provide energy efficiency in A, B, and C classes. In addition, machine learning, a sub-branch of artificial intelligence, can be customized and work on the sensor. Tensorflow Light, which is a machine learning platform Tensorflow's customized version, is applied in embedded systems is one of them (8).

3.1. Artificial Intelligence

Artificial intelligence is widely used in supply chain and logistics management as well as in every area of our lives. When we look at the historical development process of artificial intelligence, we see that it goes back to the 1950s. There has always been an open question of whether machines can think like humans. From this point of view, we can define AI as systems or machines that mimic human thinking and behavior (9).

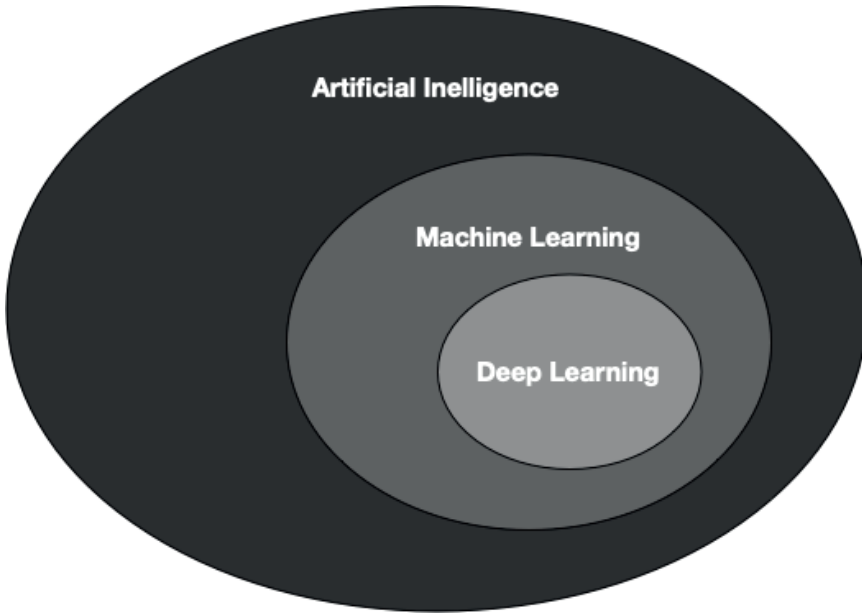


Figure 3. Artificial Intelligence, Machine Learning and Deep Learning

Artificial Intelligence is a significant discipline; some sub-concepts and topics make it up. Machine Learning and Deep Learning are at the forefront of topics widely used in today's applications. Figure 3 shows the relationship between Artificial Intelligence, Machine Learning, and Deep Learning. Deep Learning is a sub-field of Machine Learning, while Machine Learning is a sub-field of Artificial Intelligence. Machine Learning algorithms are basically grouped under two main headings supervised and unsupervised learning. While supervised learning consists of labeled datasets, unsupervised learning does not have labels.

Artificial intelligence is used in most IoT-based supply chain and logistics management aspects, from sensors to enterprise resource planning (ERP) applications on the cloud. Using TensorFlow Lite on microcontrollers with limited resources makes it possible to develop smart applications with the information obtained from the sensors. For example, an IoT module with an acceleration sensor can infer whether the object is moving or not, and if it is in

motion, it can detect which axis it is moving. While performing these operations, the available resources will be sufficient without using any cloud or server connection. Another application example, it is possible to detect at what speed objects are moving. Navigation systems can be designed for indoor warehouses. Thanks to sensors customized for Supply Chain and Logistics, objects that need to be kept under specific temperatures and pressure can be monitored by data reading and machine learning analysis of these sensors.

Thanks to the machine learning technique used in devices with limited capacity, it is possible to design energy-efficient devices without continuous data transfer to cloud servers. This approach will minimize costs by extending the life of battery-powered mobile sensors.

Expanding Machine Learning and Artificial Intelligence applications with satellite-assisted systems is possible. For this, the satellite-oriented internet of things is technologically emerging. It offers the opportunity to monitor objects worldwide with its satellite-based communication tools. Furthermore, with Artificial Intelligence, it is possible to make high-cost satellite communication more efficient by optimizing data transmission periods (10).

The variety and level of data production have increased with the Internet of Things sensors. Administrative tools can be developed by analyzing this data with Machine Learning. This way, predictions about the future can be made based on ML algorithms. For example, a business can operate without stock, and the rate of change of stocks and the products that will be needed in the future can be ordered by estimating. In this way, stock holding costs will be minimized. These analyzes are usually gathered under data analytics or big data analytics.

3.2. Big Data Analytics

In the past, the source of the data was humans. Today, not only humans but also machines produce data. The produced data can be structured or unstructured from database tables, the working sound of a machine, a picture of an object, or a video stream of a security camera. The change in the size, shape, and speed of data has caused data analysis with classical methods to be insufficient and has led to the development of big data analytics. Big Data Analytics are methods in which data are analyzed with machine learning methods apart from classical methods. In order to consider data analysis to be big data, the data must have volume, velocity, value, veracity, and a variety of properties known as 5V (11).

With big data analytics, how data is held and processed has also changed direction. Now, there is software such as Apache Hadoop and Apache Spark, where distributed structures replace central data centers.

Today, in supply chain applications, data is produced from many sources, such as sensors,

people, trade sites, and logistics. Issues such as analysis and future forecasting in supply chain management, which are quite complex in nature, can be resolved with big data analytics.

In Supply Chain management, in addition to known parameters such as capacity, demand, and cost, there are also uncertain and risky variables. Many computational methods and statistical tools have been developed to make future predictions in uncertain environments. When the data parameters included in the supply chain management are examined, It contains information such as stores, customers, products, sales, orders, shipping, and delivery. Reducing costs can be reduced by analyzing these data in detail and making estimations. In addition, it is possible to forecast the future in the fields of risk and quality management, product planning and control, logistics planning, and demand management by using big data analytics. Therefore, supply chain demand estimation is an important issue, and Artificial Neural Networks, Regression, Time Series estimation, Support Vector Machines, and Decision trees are among the most frequently used methods (12).

Another technology where data is tracked more effectively in logistics and supply chain management applications is blockchain.

3.3. Blockchain

The first idea about digital ledger and the blockchain dates back to 1990s. However, the technology has become popular with its financial application, the cryptocurrency Bitcoin (13). So when a decentralized trust-based cryptocurrency network was introduced, it caught awareness and started appearing in different applications from education to health, finance to logistics, and supply chain.

The idea of a distributed system is it does not require central management. In blockchain networks, tasks on the network, such as transactions, work peer to peer. Both systems have their advantages and disadvantages. The main disadvantage of a distributed system is performance. The centralized architectures have faster operation times since everything is coordinated and validated through a central system. On the other hand, operations on distributed systems can take a lot of work and time. Figure 4 represents centralized and decentralized network architectures.

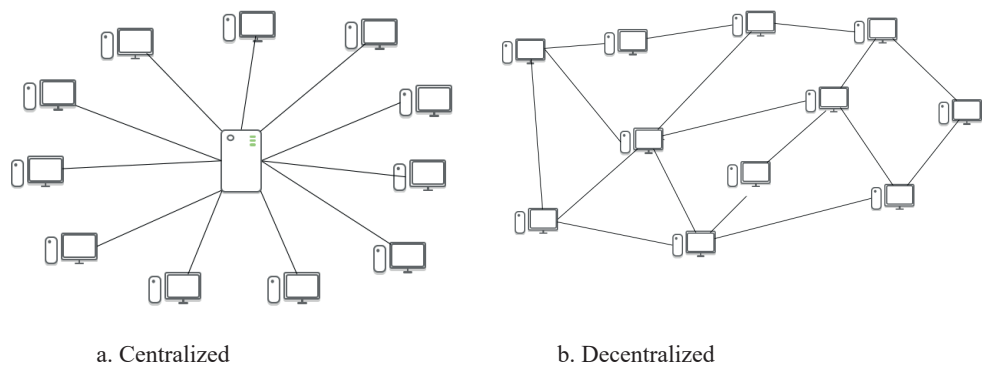


Figure 4: Centralized and decentralized architectures

Blockchain depends on secure encryption and consensus methods that allow data to be persistent, correct, and unmodifiable in the network. A record represents the single traction, which is called a block (Figure 5). Blocks knit together to construct the blockchain. Blockchain is a distributed database system that does not require central control. Whenever a transaction is triggered, the operation is distributed to all nodes over the blockchain network. Therefore, it is necessary to validate if the operation is valid. If the action is valid, it is agreed to be persistent in the database, which is the ledger. Therefore, an agreement mechanism is essential to check if the current action is correct; it is called Proof of Work.

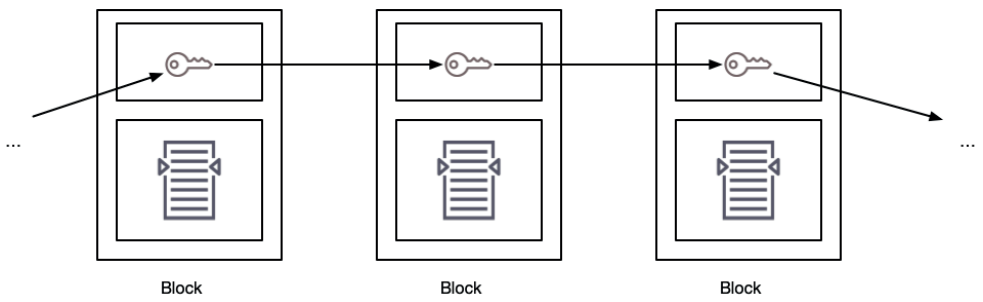


Figure 5. Blocks and blockchain

Once a transaction is verified and added to the blockchain network, any party can no longer modify it. In this way, it is possible to develop transparent applications. As a result, a more transparent process can be carried out with the blockchain infrastructure in all processes and transactions of the supply chain management process, from order placing to delivery.

Blockchain applications can be used in private or public network infrastructures. A logistics process or firm may privately set up its infrastructure or maintain its services from a public

network infrastructure. If we look at the prominent blockchain approaches, blockchain can be used in authentication and authorization processes. Authentication allows the authorization of both human and agent applications with artificial intelligence and the strategies to be automated. Moving product tracking and logistics processes to a different dimension with Blockchain infrastructure is possible. In particular, all movements of an object can be recorded without compromising information security and can be followed transparently by various authorities.

Using the trust mechanisms provided by blockchain structures without central management in logistics processes with many stakeholders is possible. In addition, with the smart contract applications that come with the blockchain, supply chain processes can be reduced to more automated and less human-independent processes that are subject to fewer errors.

3.4. Digital Twins

Digital twins are virtual representations of physical systems that enable real-time analysis, simulation, and management. We can consider the three main layers of the digital twin architecture. These layers are the data layer, virtualization layer, and service layer. Data obtained from different physical layers are consolidated in the data layer, and data communication with other layers is provided. IoT sensors or third-party system data are examples of these data sources. The virtualization layer is the process that includes the virtual representation of a physical object. In this layer, a physical object or a process is designed one-to-one in the virtual environment. On the other hand, the service layer contains the tools through which the end users benefit from the services (14).

The most important feature distinguishing digital twin technology from other simulations is real-time analysis with two-way data flow. An analysis consisting of live data can be made on a digital twin that is fed instantly from a physical entity or process. It is possible to perform simulations on different scenarios. In this approach, accurate replication of the physical twin is modeled. Thus, the digital twin is a realistic replica of a physical twin. Digital Twin systems do not use the abstraction approach of classical simulations. Instead, they perform instant analysis on a more realistic model. Another difference is that the physical twin can be updated due to analysis. This enables the instantaneous transfer of the obtained configuration space through an analysis or a simulation of the physical system. In this way, a bidirectional data transfer is guaranteed.

Digital twin technologies, which stand out in the field of supply chain management and logistics and are planned to take place in all sectors in the future, are based on the real-time replication of a physical object or process in virtual environments. This technology offers the

opportunity to perform analysis, simulation, configuration, and management activities with digital twins, which cannot be done with classical methods in a virtual replica of a real system. Using real system data in real-time enables us to perform what-if simulation analysis, future predictions, and predictive maintenance with digital twins(14).

Digital Twin technologies play a role in the execution of many monitoring and administrative activities in the field of logistics. In addition, it plays a role in many areas of usage, such as optimization of packaging and collection processes, smart management of container fleets, 3D modeling of containers, and repair or scrapping of products to be used in the fleet based on historical data (15).

4. Conclusion

With rapid technological developments and low hardware costs, all electronic devices and sensors have started to serve by connecting to the internet. In this way, the transformation of the Internet of Things began and shifted to the Industry of the Industrial Internet of Things (IIoT). This study examines IoT's data transmission structure, application protocols, and usage areas in Supply Chain Management and Logistics applications. In addition, the application areas of artificial intelligence, big data, blockchain and digital twin technologies and their contributions to Supply Chain and Logistics management were evaluated.

The usage area of Digital Twin technologies has become widespread due to the rapid development of wireless communication technologies, the development of wireless and fast broadband networks, and the acceleration of studies in this field. Studies show the logistics and supply chain processes of Digital Twins, and Digital Twins will be used more widely context in the future. In this context, it will be necessary to create realistic models of logistics processes using Digital Twins. Especially in high-volume object tracking processes, the Digital Twin of each object will increase its spatial and computational complexity. It is an open research problem to develop more efficient methods with research in this field. In addition, the bidirectional data flow and the configuration of the physical object hosted by the Digital Twins will also bring cyber security threats. Seizing a digital copy of an industry's physical assets and processes and unauthorized transactions therein will raise serious security threats. Research on the subject is still underway to ensure the safe and secure integration of Digital Twins in logistics and supply chain management.

Blockchain applications still need to be actively used in Digital Twins and logistics and supply chain processes. The security measures that this approach will bring, the delays in blockchain consensus mechanisms, and whether the Digital Twins will solve the needs of instant data analysis or the development of alternative consensus mechanisms will be an open issue.

References

1. Shancang L, Li Da X, Shanshan Z. The internet of things: a survey. *Inf Syst Front*. 2015;17(2):243-59.
2. Sethi P, Sarangi SR. Internet of Things: Architectures, Protocols, and Applications. *J Elec Comput Eng* 2017;2017:9324035.
3. Khanna A, Kaur S. Internet of Things (IoT), Applications and Challenges: A Comprehensive Review. *Wirel Pers Commun* 2020;114(2):1687-762.
4. Ertürk MA, Aydın MA, Büyükakkaşlar MT, Evirgen H. A Survey on LoRaWAN Architecture, Protocol and Technologies. *Future Internet* 2019;11(10):216.
5. Johnson D, Ketel M. IoT: Application Protocols and Security. *Int J Comput Netw Inf Sec* 2019;11(4):1-8.
6. Kebande VR. Industrial internet of things (IIoT) forensics: The forgotten concept in the race towards industry 4.0. *Forensic Sci Int: Rep* 2022;5:100257.
7. Abdel-Basset M, Manogaran G, Mohamed M. Internet of Things (IoT) and its impact on supply chain: A framework for building smart, secure and efficient systems. *Future Gener Comput Syst* 2018;86:614-28.
8. Lite T. Machine Learning for Mobile and Edge Devices 2023 [Available from: <https://www.tensorflow.org/lite>].
9. Stuart R, Peter N. Artificial Intelligence: A Modern Approach: Pearson Publications; 2016.
10. Centenaro M, Costa CE, Granelli F, Sacchi C, Vangelista L. A Survey on Technologies, Standards and Open Challenges in Satellite IoT. *IEEE Commun Surv Tutor* 2021;23(3):1693-720.
11. Ishwarappa, Anuradha J. A Brief Introduction on Big Data 5Vs Characteristics and Hadoop Technology. *Procedia Comput Sci* 2015;48:319-24.
12. Seyedan M, Mafakheri F. Predictive big data analytics for supply chain demand forecasting: methods, applications, and research opportunities. *J Big Data* 2020;7(1):53.
13. Nakamoto S. Bitcoin: A peer-to-Peer Electronic Cash [Available from: <https://bitcoin.org/bitcoin.pdf>].
14. Tao F, Zhang H, Liu A, Nee AYC. Digital Twin in Industry: State-of-the-Art. *IEEE Trans Ind Inform* 2019;15(4):2405-15.
15. DHL. Digital Twin in Logistics, A DHL perspective on the impact of digital twin on the logistics industry [Available from: <https://www.dhl.com/content/dam/dhl/global/csi/documents/pdf/csi-ltr6-resources-digital-twins-in-logistics-dhl-trend-report.pdf>].

